GRADE 7 ACCELERATED · UNIT 6 · LESSON 3

Pythagorean Theorem

Independent Set 6.03 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. A right triangle has legs of 12 m and 5 m. How long is the hypotenuse?

13 m

144 + 25 = 169 = 13^2.

2. A right triangle has legs of 3 ft and 8 ft. How long is the hypotenuse, exactly and to the nearest tenth?

$\sqrt{73}$ ft ≈ 8.5 ft

9 + 64 = 73.

3. A right triangle has a hypotenuse of 39 cm and one leg of 36 cm. How long is the other leg?

15 cm

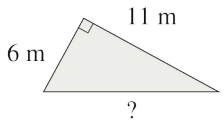
1,521 − 1,296 = 225 = 15^2.

4. A right triangle has a hypotenuse of 11 inches and one leg of 4 inches. How long is the other leg, exactly and to the nearest tenth?

$\sqrt{105}$ in ≈ 10.2 in

121 − 16 = 105.

5. A right triangle has legs of 6 m and 11 m (drawn turned on its side on the paper). How long is its hypotenuse, to the nearest tenth?



$\sqrt{157}$ m ≈ 12.5 m

36 + 121 = 157; c is across from the square corner.

6. A right triangle has a hypotenuse of 25 yd and one leg of 24 yd. The other leg is x. Write the equation that finds x, then find x.

$24^2 + x^2 = 25^2$; x = 7 yd

625 − 576 = 49 = 7^2.